

Student Number

**YEAR 12 ASSESSMENT BLOCK  
SEMESTER 1**

### Section I - Multiple Choice

Write your student number at the top of this page.

Remove this sheet.

Select the alternative A, B C or D that best answers the question. Fill in the response oval completely.

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If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

*,correct*

|    |                         |                         |                         |                         |     |                         |                         |                         |                         |
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| 1. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D | 6.  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 2. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D | 7.  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 3. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D | 8.  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 4. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D | 9.  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 5. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D | 10. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |



# Barker

College

AXD AYG\*  
 ESP LMD  
 JZT RJW\*  
 ALY RAS  
 ARP

Student Number

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Friday, 19<sup>th</sup> June 2020  
 12 PM  
 120 minutes

## Year 12 MATHEMATICS ADVANCED Assessment Task #2

Discrete Probability Distributions, Graphs & Equations,  
 Curve Sketching using the Derivative, Integration,  
 The Exponential & Logarithmic Functions,  
 The Trigonometric Functions

145 copies

### General Instructions

- Write using blue or black pen
- NESA approved calculators may be used
- Answer each question in the space provided.
- A reference sheet is provided separately
- In Questions 11-15, show relevant mathematical reasoning and/or calculations
- Diagrams are NOT to scale

|                     | Section         | Marks       |
|---------------------|-----------------|-------------|
| <b>I</b>            | Multiple Choice | / 10        |
| <b>II</b>           | Question 11     | / 14        |
|                     | Question 12     | / 14        |
|                     | Question 13     | / 14        |
|                     | Question 14     | / 15        |
|                     | Question 15     | / 13        |
| <b>Total Marks:</b> |                 | <b>/ 80</b> |

### Section I

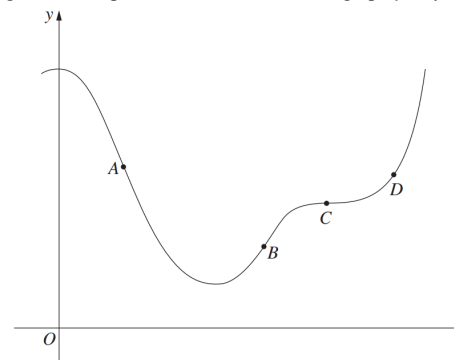
10 Marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

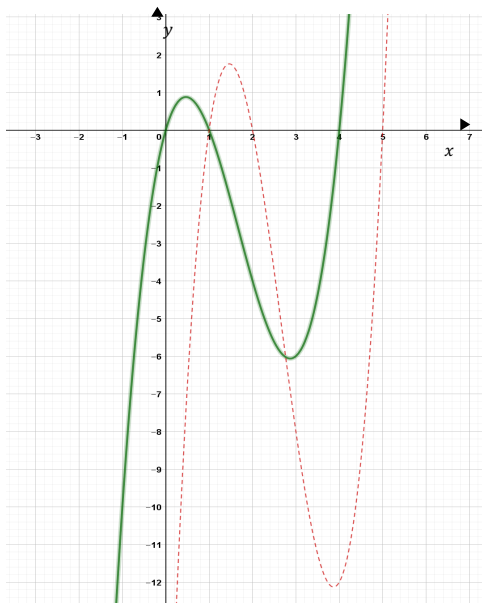
- The domain of  $f(x) = \ln(x - 1)$  in interval notation is given by:
  - $(-\infty, \infty)$
  - $(-1, \infty)$
  - $(0, \infty)$
  - $(1, \infty)$
- Using the log laws, fully simplify:  $12\log_a(\sqrt{a})$ 
  - 6
  - 12
  - $6\log_a(a)$
  - $\sqrt{12}$
- The diagram shows points  $A, B, C$  and  $D$  on the graph  $y = f(x)$ .



At which point is  $f'(x) > 0$  and  $f''(x) = 0$

- A
- B
- C
- D

- 4 The graph of  $y = f(x)$  is shown as a solid line.



Which of the following is a possible equation for the dotted line?

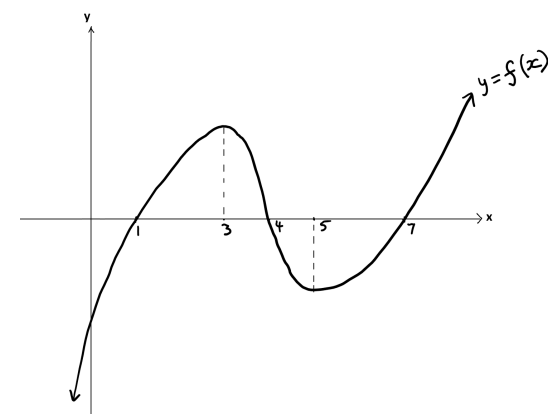
- (A)  $y = -2f(x)$   
 (B)  $y = 2f(2x - 1)$   
 (C)  $y = -2f(x - 1)$   
 (D)  $y = 2f(x - 1)$
- 5 The gradient function of  $y = x^3 e^{2x}$  is given by:
- (A)  $3x^2 e^{2x}$   
 (B)  $6x^2 e^{2x}$   
 (C)  $3x^2 e^{2x} + x^3 e^{2x}$   
 (D)  $x^2 e^{2x}(2x + 3)$

- 6 The equation of the tangent to  $y = \cos x$  at the point  $(2\pi, 1)$  is given by:

- (A)  $y = -\sin(x) + 1$   
 (B)  $y = x - 2\pi$   
 (C)  $y = -\sin(x) - 2\pi$   
 (D)  $y = 1$

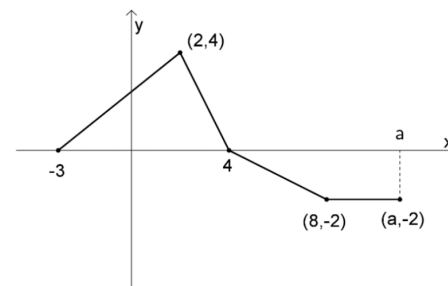
- 7 Using the sketch provided, determine when  $f'(x) < 0$  in interval notation.

- (A)  $[1, 4] \cup [7, \infty)$   
 (B)  $(-\infty, 1) \cup (4, 7)$   
 (C)  $(3, 5)$   
 (D)  $(-\infty, 3) \cup (5, \infty)$



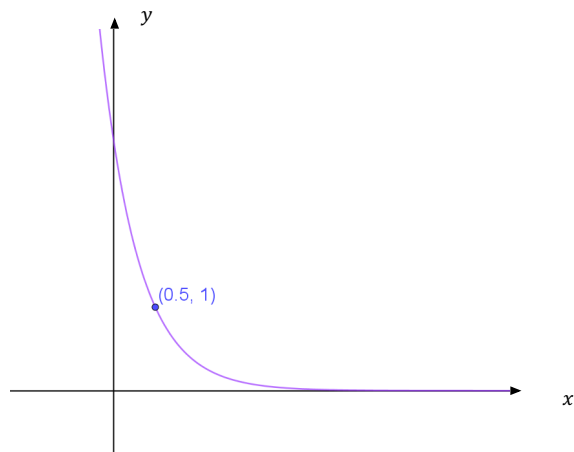
- 8 Using the graph of  $y = f(x)$ , determine the value of  $a$  which satisfies the condition:

$$\int_{-3}^a f(x) dx = 0$$



- (A) 5  
 (B) 9.5  
 (C) 13  
 (D) 18

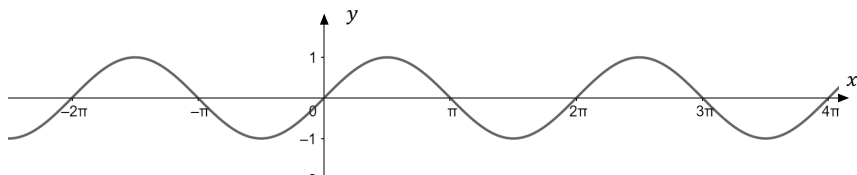
- 9 The graph below shows a translation and dilation of the curve  $y = 3^{-x}$ . Which equation below matches the graph?



- (A)  $y = 3^{-2x+1}$   
 (B)  $y = 3^{-2(x-1)}$   
 (C)  $y = 3^{-2x-1}$   
 (D)  $y = 3^{-2(x+1)}$

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- 10 The graph of  $y = \sin x$  is shown below.



Determine the maximum possible value of  $\int_a^b \sin x \, dx$ , where  $a$  and  $b$  are real numbers.

- (A) 0  
 (B) 2  
 (C)  $\infty$   
 (D)  $\pi/2$

## Section II

70 Marks

Attempt Questions 11-15

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate space provided.

In Questions 11-15, your responses should include relevant mathematical reasoning and/or calculations.

## Question 11 (14 marks)

(a) Find these derivatives

$$(\alpha) \quad \frac{d}{dx}(x^2 - 5)^6 \quad 1$$

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$$(\beta) \quad \frac{d}{dx} \frac{\ln(3x)}{e^x} \quad 2$$

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$$(\gamma) \quad \frac{d}{dx} 2^x e^x \quad 2$$

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(b) Find the indefinite integrals

$$(\alpha) \quad \int 5 - x^3 dx \quad 1$$

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## Question 11 (continued)

$$(\beta) \quad \int (2x-1)^5 dx \quad 2$$

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$$(\gamma) \quad \int \cos(3x-1) dx \quad 2$$

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(c) Evaluate the following definite integrals

$$(\alpha) \quad \int_0^1 e^{5x} - 1 dx \quad 2$$

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$$(\beta) \quad \int_{-2}^1 \frac{2}{x+3} dx \quad 2$$

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**Question 12** (14 marks)

(a) Consider the probability distribution below.

2

- (i) Show that  $p = \frac{1}{12}$  and hence complete the discrete probability distribution table, assuming no other values of  $x$  are possible.

| $x$        | 0   | 1              | 2    | 3    | 4             | 5             | 6   | 7              |
|------------|-----|----------------|------|------|---------------|---------------|-----|----------------|
| $P(X = x)$ | $p$ | $\frac{1}{12}$ | $2p$ | $2p$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $p$ | $\frac{1}{12}$ |

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- (ii) Find  $E(X)$ .

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- (iii) Calculate the  $Var(X)$

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- (iv) Calculate the standard deviation, to 2 decimal places.

1

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**Question 12** (continued)

(b)

- (i) Sketch the curve  $f(x) = -x(x - 1)(x + 3)$ , showing  $x$  and  $y$  intercepts, in the space below.

2

- (ii) Hence or otherwise, solve  $f(x) \leq 0$ , giving your answer in interval notation.

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- (c) For a certain curve  $y = f(x)$ ,  $f''(x) = 6x - 1$ .

3

Find the equation of the curve given that it passes through the point  $(1, -2)$  and has a gradient of  $-3$  at that point.

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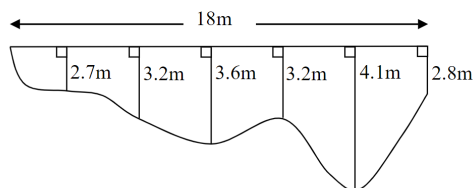
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(a) A surveyor measures the depth of a river at equal intervals across its width.

3

NOT TO  
SCALE



Using the trapezoidal rule with 7 function values, find the approximate cross – sectional area of the river.

[illegible]

2

[illegible]

(c) Consider the function  $y = x^3 - 3x^2 - 9x + 1$

4

[illegible]

**2**

[illegible]

**Question 13** (continued)

- (iii) Sketch the graph  $y = x^3 - 3x^2 - 9x + 1$  in the interval  $[-3, 6]$  in the space below, showing all important features. (No need to find the  $x$ -intercepts) **2**

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- (iv) Find the maximum value of the function in the domain  $[-3, 6]$ . **1**

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**Question 14** (15 marks)

- (a) The table below is a probability distribution table for a random variable  $X$  with  $\mu = 23$ .

|            |     |     |     |
|------------|-----|-----|-----|
| $x$        | 10  | $a$ | 30  |
| $P(X = x)$ | 0.2 | $b$ | 0.3 |

Find the values of  $a$  and  $b$ .

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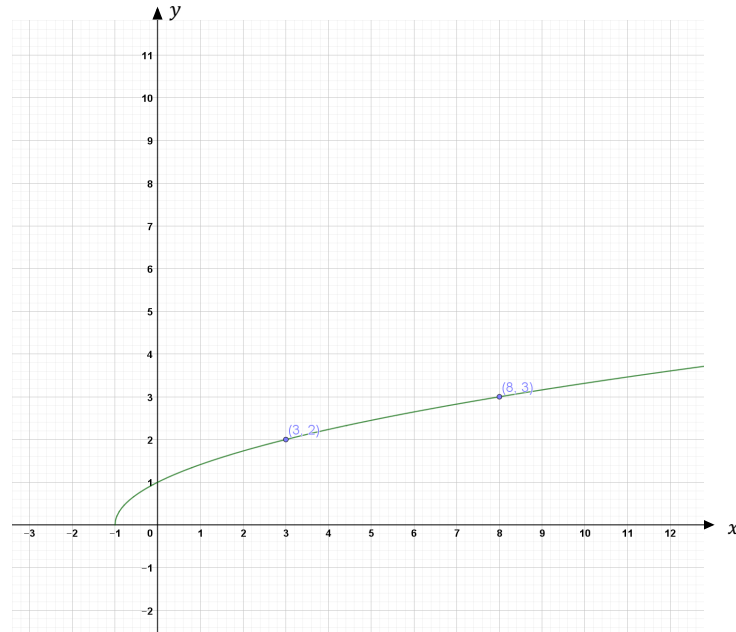
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- (b) The graph of  $f(x) = \sqrt{x+1}$  is shown below.

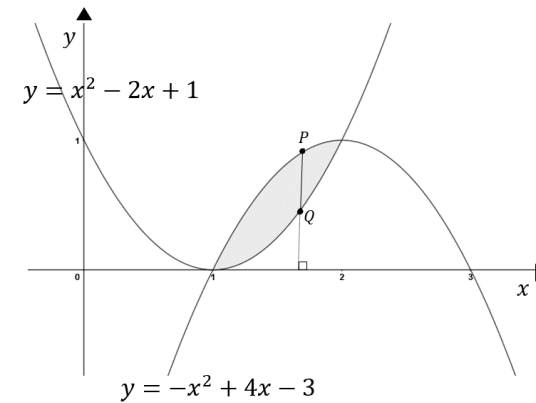
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Use dilations to sketch  $y = 3f(2x)$  on the same set of axes, labelling the image points of  $(3, 2)$  and  $(8, 3)$  and the y-intercept (points on the new graph which correspond to these points).



**Question 14** (continued)

- (c) The diagram shows the curves  $y = x^2 - 2x + 1$  and  $y = -x^2 + 4x - 3$



- (i) Show that the curves intersect at the points where  $x = 1$  and  $x = 2$

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- (ii) Hence, solve  $x^2 - 2x + 1 < -x^2 + 4x - 3$ .

1

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- (iii) If the length of  $PQ$  is  $l$  units, show that  $l = -2x^2 + 6x - 4$ .

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**Question 14 (continued)**

- (iv) Hence find the maximum length of  $PQ$  for  $[1, 2]$ . **3**

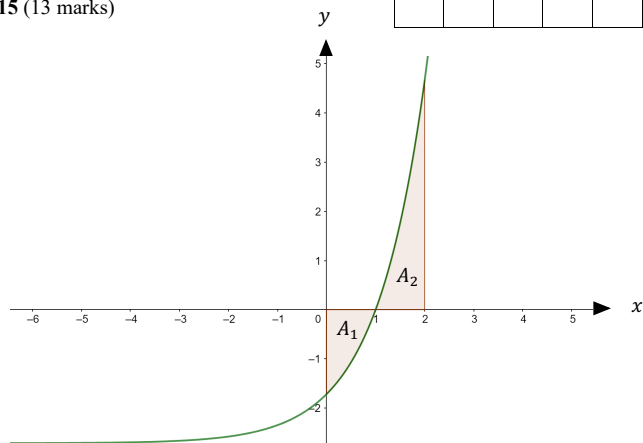
[illegible]

- (v) Find the area of the shaded region enclosed between the two curves. 3

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(a)



(i) Show that the  $x$ -intercept is at  $(1, 0)$ .

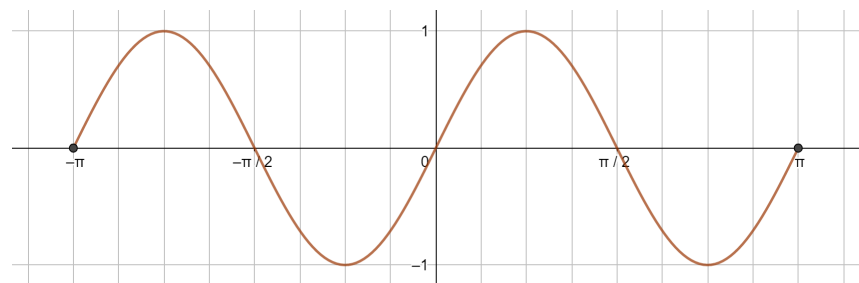
1

(ii) Hence, or otherwise, find the total shaded area ( $A_1 + A_2$ ), in the diagram above.

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Give your answer in exact form.

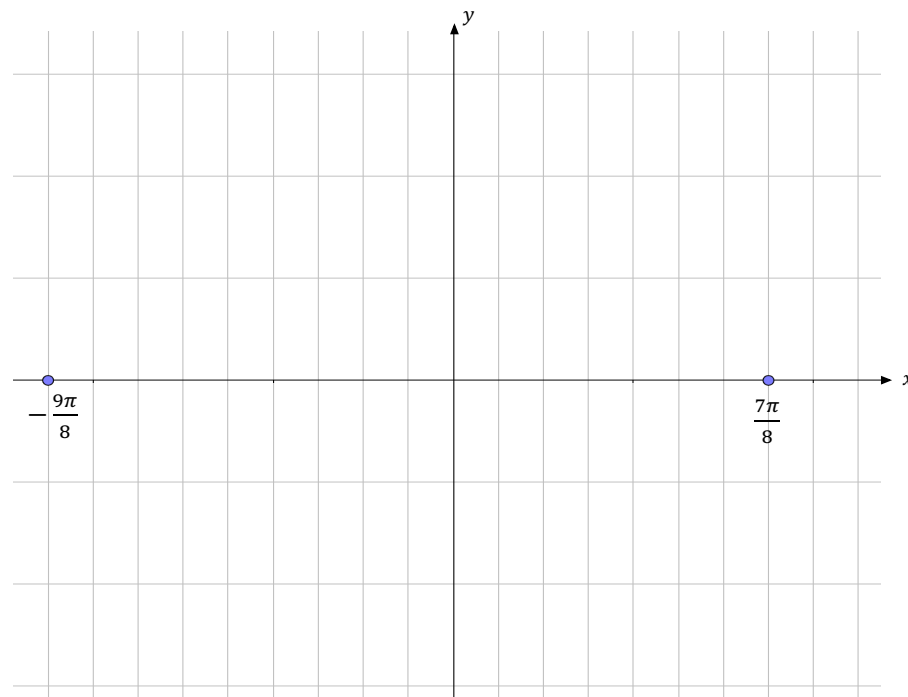
(b) The trigonometric graph below shows the function  $y = \sin 2x$  in the interval  $[-\pi, \pi]$



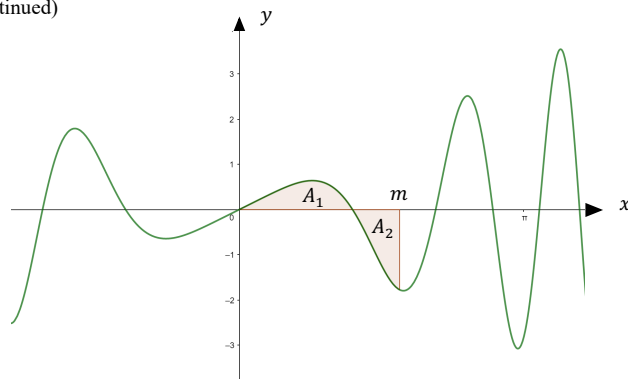
Hence, or otherwise sketch  $y = 3 \sin \left[ 2 \left( x + \frac{\pi}{8} \right) \right]$  in the interval  $\left[ -\frac{9\pi}{8}, \frac{7\pi}{8} \right]$  on the axes below.

Label all the intercepts and mark a scale on the  $y$  axis.

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(c)



The diagram above shows part of the curve  $y = x\cos(x^2)$ .

- (i) Find  $\frac{d}{dx} \sin(x^2)$ . **1**

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- (ii) Hence find  $\int x \cos(x^2) dx$ . 2

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- (iii) Given that  $\text{area } A_1 = \text{area } A_2$ , find the value of  $m$  marked on the diagram. 3

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**END OF ASSESSMENT**

1. D.  $f'(1,0)$
  2. A.  $12 \log_a \sqrt{a} = 6$
  3. B. Positive Gradient and POI
  4. D.  $y = 2f(x-1)$
  5. D.  $u = x^3 \quad v = e^{2x}$   
 $u' = 3x^2 \quad v' = 2e^{2x}$   
 $y' = x^2 e^{2x} (2x+3)$
  6. D.  $y' = -\sin x$   
 $x = 2\pi \quad m = 0$   
 $\therefore y = 1$
  7. C.  $(3,5)$
  8. C.  $A = 14 - 4 + 10$   
 $\therefore a = 13$
  9. A.  $y = 3^{-2x+1}$
  10. B.  $\int_0^{\pi} \sin x dx = -\cos x \Big|_0^{\pi}$   
 $= 2$
  11. a) (A)  $= 6(x^2-5)^5 \times 2x$   
 $= 12x(x^2-5)^5$   
 (B)  $u = \ln 3x \quad v = e^x$   
 $u' = \frac{1}{x} \quad v' = e^x$   
 $y' = \frac{\frac{e^x}{x} - e^x \ln 3x}{e^{2x}}$   
 $= \frac{1-x \ln 3x}{x e^{2x}}$   
 (C)  $u = 2^x \quad v = e^x$   
 $u' = \ln 2^x \quad v' = e^x$   
 $y' = 2x e^x (\ln 2 + 1)$   
 b)  $y = 5x - \frac{x^4}{4} + C$   
 (A)  $= \frac{(2x-1)}{6} + C$   
 $= \frac{(2x-1)^6}{12} + C$
  12. a) (i)  $6p + \frac{1}{2} = 1$   
 $6p = \frac{1}{2}$   
 $p = \frac{1}{12}$   
 (ii)  $E(x) = 0 + \frac{1}{12} + \frac{2}{6} + \frac{3}{6} + \frac{4}{6} + \frac{5}{6} + \frac{6}{12} + \frac{7}{12}$   
 $= 3.5$   
 (iii)  $= E(x^2) - \mu^2$   
 $= \frac{97}{6} - 3.5^2$   
 $= 3.92$   
 (iv)  $= \sqrt{3.92}$   
 $= 1.98$   
 b) (i)   
 (ii)  $[-3, 0] \cap [1, \infty)$
  - c)

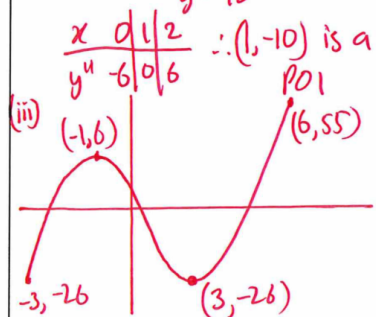
$f'(x) = 3x^2 - x + c$   
 $f'(1) = 2 + c = -3$   
 $c = -5$   
 $\therefore f'(x) = 3x^2 - x - 5$   
 $f(x) = x^3 - \frac{x^2}{2} - 5x + d$   
 $f(1) = 1 - \frac{1}{2} - 5 + d = -2$   
 $d = 2.5$   
 $\therefore f(x) = x^3 - \frac{x^2}{2} - 5x + \frac{5}{2}$   
 13. a) Area  $\div \frac{3}{2}(0 + 2.8 + 2(16.8))$   
 $\div 54.6 \text{ m}^2$

b)  $f'(x) = 6x^2 + 2x > 0$   
 $2x(3x+1) > 0$   
 $x < -\frac{1}{3}, x > 0$

c) (i)  $y' = 3x^2 - 6x - 9 = 0$   
 $(x-3)(x+1) = 0$   
 $x = 3, -1$   
 $y = -26, 6$

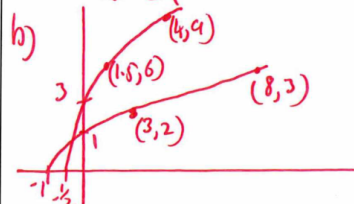
$y'' = 6x - 6$   
 $y''(3) = 12 \therefore (3, -26) \text{ max}$   
 $y''(-1) = -12 \therefore (-1, 6) \text{ min}$

(ii)  $y'' = 6x - 6 \geq 0$   
 $x = 1$   
 $y = -10$



(iv)  $x = 6, y = 55$   
 14. a)  $0.2 + b + 0.3 = 1$   
 $b = 0.5$

$23 = 2 + \frac{a}{2} + 9$   
 $a = 24$



c) (i)  $x^2 - 2x + 1 = -x^2 + 4x - 3$   
 $(x-2)(x-1) = 0$   
 $\therefore x = 1, 2$

(ii)  $1 < x < 2$

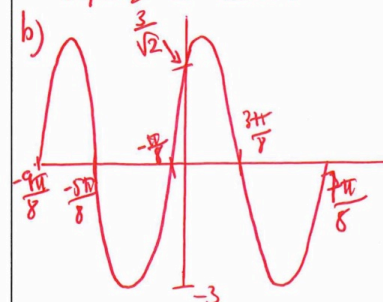
(iii)  $l = \text{top} - \text{bottom}$   
 $= -2x^2 + 6x - 4$

(iv)  $\frac{dl}{dx} = -4x + 6 = 0$   
 $x = \frac{3}{2}$   
 $\frac{d^2l}{dx^2} = -4 \therefore x = \frac{3}{2} \text{ max}$   
 $\text{Max} = -2\left(\frac{3}{2}\right)^2 + 6\left(\frac{3}{2}\right) - 4$   
 $= \frac{1}{2} \text{ unit}$

(v)  $A = \int_1^2 (-2x^2 + 6x - 4) dx$   
 $= \left[-\frac{2x^3}{3} + 3x^2 - 4x\right]_1^2$   
 $= -\frac{16}{3} + 12 - 8 - \left(-\frac{2}{3} + 3 - 4\right)$   
 $= \frac{1}{3} \text{ u}^2$

15 a) (i)  $e^x - e = 0$   
 $x = 1$   
 $\therefore 1, 0 \text{ is } x \text{ int}$   
 (ii)  $A_1 = \left| \int_0^1 e^x - e dx \right|$   
 $= |e^x - ex|_0^1$   
 $= |e - e - 1|$   
 $= 1$   
 $A_2 = e^x - ex$   
 $= e^2 - 2e$

$A_1 + A_2 = e^2 - 2e + 1$



c) (i)  $2x \cos x^2$   
 (ii)  $\int x \cos x^2 dx = \frac{1}{2} \int 2x \cos x^2 dx$   
 $= \frac{1}{2} \sin x^2 + C$

(iii)  $\int_0^m x \cos x^2 dx = 0$   
 $\therefore \frac{\sin x^2}{2} \Big|_0^m = 0$   
 $\sin m^2 = 0$   
 $\therefore m^2 = 0, \pi, 2\pi, \dots$   
 $m = \sqrt{\pi}$   
 1st value